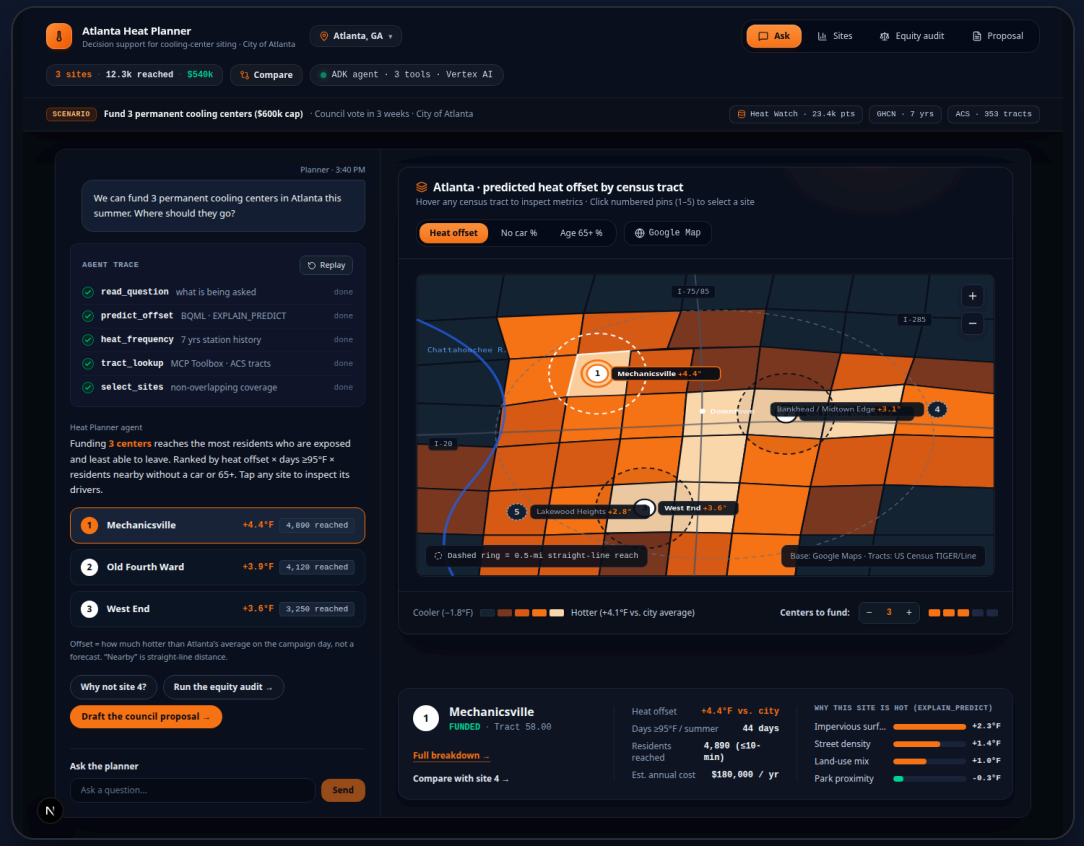
Three Permanent Cooling Centers, Sited Where Heat Hurts Most

The Municipal Decision Scenario (Atlanta, GA)

- **Who Is Asking:** Dana, Mayor's Office of Sustainability & Resilience (3:40 PM Tuesday; City Council votes in 3 weeks on the FY26 capital budget).
- **The Mandate:** Allocate **$600,000** to fund **3 permanent cooling centers** ($180k/yr each) across 353 Atlanta census tracts.
- **The Microclimate Inequality:** Airport forecasts report one city temperature, but **23,386 street-level Heat Watch readings** (mean 82.06°F) show blocks 2 miles apart differ by up to **13.5°F** (-5.79°F to +7.70°F offset).
- **Why Raw Sensors Fail Council:** Siting on peak heat alone puts centers in rail yards or clusters all 3 in one district; siting on politics leaves car-free seniors unserved.
- **Our Solution:** A Google Cloud ADK + BigQuery ML spatial decision assistant that predicts street-level heat offsets (°F above city mean), explains physical drivers, enforces non-overlapping catchments, and audits equity.

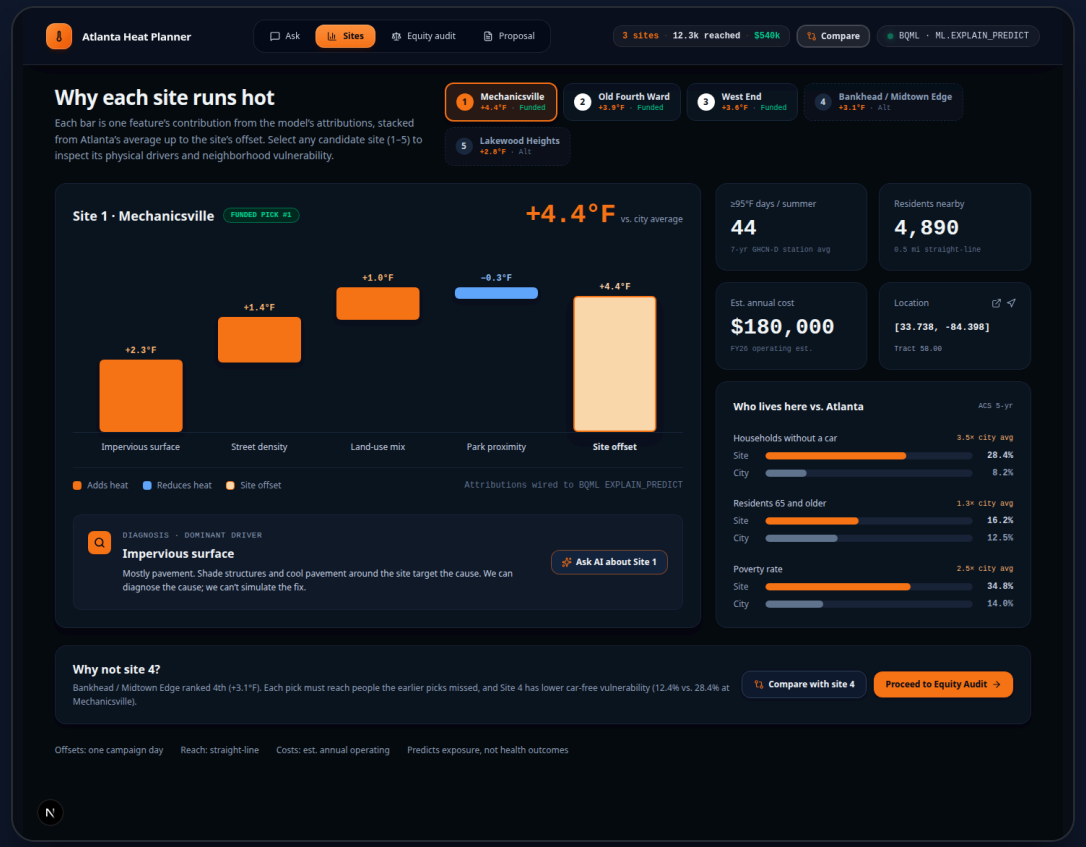
> ***Speaker Note:*** *"We chose Atlanta on the 'This Budget Cycle' track. Our planner has $600k to site 3 permanent cooling centers before Council votes in 3 weeks. We built an AI assistant that predicts microclimate heat offsets, proves who benefits, and survives a contested City Council hearing."*



The Answer: 3 Non-Overlapping Sites + Why Site #4 Was Displaced

3 Funded Picks ($540k / $600k Cap · 12,260 Reached in 0.5 mi)

- **Offset, Never Raw Temperature:** Ranks 5,618 candidate grid points (~250 m) by **Predicted Offset (+°F vs. 82.06°F city mean) × 7-yr GHCN Hot Days (≥95°F) × Walkable Vulnerability** with greedy **≥2.0 km spacing**.
- **Site #1 · Mechanicsville (Tract 58.00 / 76.03): +4.4°F vs. city mean** · 44 days/yr ≥95°F · 4,890 residents in 0.5 mi (28.4%–56.4% no-car, 16.2% 65+). `ML.EXPLAIN_PREDICT` drivers: Impervious surface (+2.3°F), Street density (+1.4°F), Land-use mix (+1.0°F), Park proximity (-0.3°F).
- **Site #2 · Old Fourth Ward / Ashview (Tract 42.00): +3.9°F vs. city mean** · 42 days/yr ≥95°F · 4,120 residents (24.1%–46.7% no-car, 17.8% 65+). Drivers: Impervious surface (+1.9°F), Street density (+1.4°F).
- **Site #3 · West End / Summerhill (Tract 48.00): +3.6°F vs. city mean** · 41 days/yr ≥95°F · 3,250 residents (22.5%–38.0% no-car, 14.6%–24.2% 65+). Unmeasured tract predicted via BQML.
- **Why Not Site #4 (Bankhead / Midtown Edge)?** Runs +3.1°F (39 days ≥95°F) but has less than half the car-free rate (12.4% vs. 28.4% at Site #1) and reaches 2,480 vs. 4,890 residents.



> ***Speaker Note:*** *"Notice two critical things: first, we predict thermal offset above the city mean—never fake point forecasts—and decompose every site into °F physical drivers via ML.EXPLAIN_PREDICT. Second, we explicitly answer why runner-up Site 4 in Bankhead*

Our Team: Built Across Four Specialized Engineering Lanes

Rama Chintamaneni (@chintamanenirc · chintamaneni.rc@gmail.com)

CLOUD PLATFORM, ADK AGENT & MCP LEAD

- **Google ADK & Tooling:** Architected the HeatPlanner ADK agent, system instructions, ground-rule refusal gates, and Google Maps Geocoding integration.
- **Managed MCP Toolbox:** Deployed private Cloud Run MCP Toolbox (`heat-toolbox`) with parameterized BigQuery tools (`tools.yaml`) and workload identity auth.
- **Spatial Siting Engine:** Built Haversine greedy spacing (`spread_sites >= 2.0 km`) and `VizSpecBuilder` contracts across PRs #6, #7, #10, and #11.

Harika (@harika126 · harika126@gmail.com)

DATA ARCHITECTURE, SEMANTIC LAYER & ORCHESTRATION LEAD

- **4-Layer Medallion Warehouse:** Engineered `a4i_core`, `a4i_semantic`, and `a4i_ml` in BigQuery (353 tracts, 2,020 geohash-7 cells, 17,452 GHCN station-days).
- **Data Quality & Licensing:** Resolved `-9999.0` sentinels, LPAD 11-char FIPS, joined US Census TIGER & ACS 65+ (100% match), and passed 13/13 SQL validation checks.
- **Zero-Skew Feature TVF:** Built `a4i_ml.point_features(pts)` with 500 m footprint refusal gate and co-built Stage 1/2 `HeatPlannerOrchestrator`.

Faizan Chundrigar (@faizy31796 · faizy31796@gmail.com)

MACHINE LEARNING & EQUITY ANALYTICS ENGINEER

- **Leak-Free BQML Model:** Trained `a4i_ml.offset_model_v1` (`LINEAR_REG`, `enable_global_explain=TRUE`, `l2_reg=0.1`) with zero drive-day or racial features.
- **Spatial-Block Validation:** Designed `geohash-5` (~4.9 km) spatial-block holdout (1,378 train / 484 eval cells) beating baseline (MAE 1.254°F vs. 1.355°F).
- **Grid Scoring & Equity Audit:** Pre-computed 5,618 ~250 m `ML.EXPLAIN_PREDICT` points (`grid_predictions`) and authored `sql/audit/audit.sql` outcome audit.

N. Viradiya (@nviradiya · Urban Heat Team)

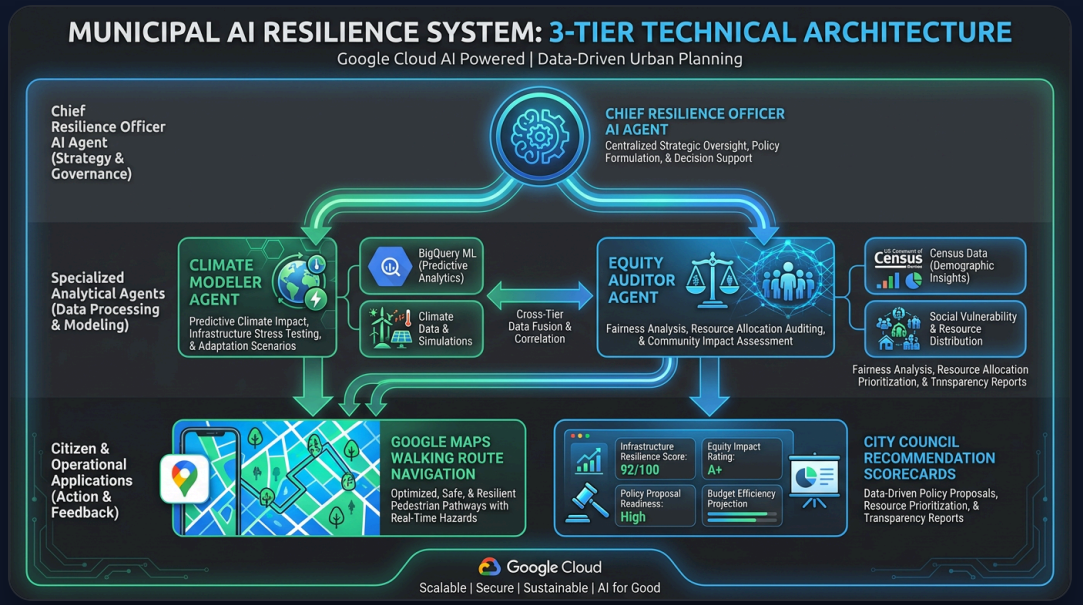
FULL-STACK AI & GEOSPATIAL UX LEAD

- **Production Next.js 16 + React 19 UI:** Built the 4-tab municipal workbench (Ask, Sites, Equity Audit, Proposal) with interactive Voronoi census tract tessellation.
- **Grounding Verifier & Live API:** Implemented `GroundingVerifier` & `FallbackRenderer` to verify every °F, %, and headcount against live BigQuery semantic data.
- **Human-in-the-Loop Governance:** Created interactive Site 1 vs. Site 4 trade-off modal, Equity Audit scatter plot, and signed Council Proposal PDF exporter.

Architecture & Why You Can Trust It: Leak-Free Spatial Holdout

End-to-End Google Cloud Stack & Validation Rigor

- **Google ADK + Gemini 2.5 Flash:** Multi-agent orchestrator calling custom Python tools (`explain_heat_offset`, `select_cooling_center_sites`, `run_equity_audit`) + `GroundingVerifier`.
- **Managed MCP Server (`heat-toolbox`):** Cloud Run MCP Toolbox for Databases exposing parameterized SQL tools over `a4i_semantic` (`dim_tract`, `site_hot_days`, `vulnerable_population`) with least-privilege IAM (`toolbox-sa`, `heat-agent-sa`).
- **Where vs. How Often Separation:** Heat Watch (23,386 readings, 100% NULL timestamps) answers spatial *WHERE*; 7-year NOAA GHCN daily history (17,452 station-days across 7 stations) answers temporal *HOW OFTEN* via `site_hot_days(offset_f)`.
- **Defeating 1 Hz Spatial Leakage:** Aggregated 23,386 points into 2,020 `geohash-7` (~150 m) cells and held out 6 whole `geohash-5` (~4.9 km) blocks (484 eval / 1,378 train). Excluded `temp_f`, `heat_index_f`, `humidity_pct`, and all race columns.
- **Verified BQML Performance:** `a4i_ml.offset_model_v1` beats predict-zero baseline on held-out spatial blocks (**MAE = 1.254°F vs. 1.355°F**; RMSE = 1.542°F vs. 1.622°F) and passes **13/13 SQL validation checks**.

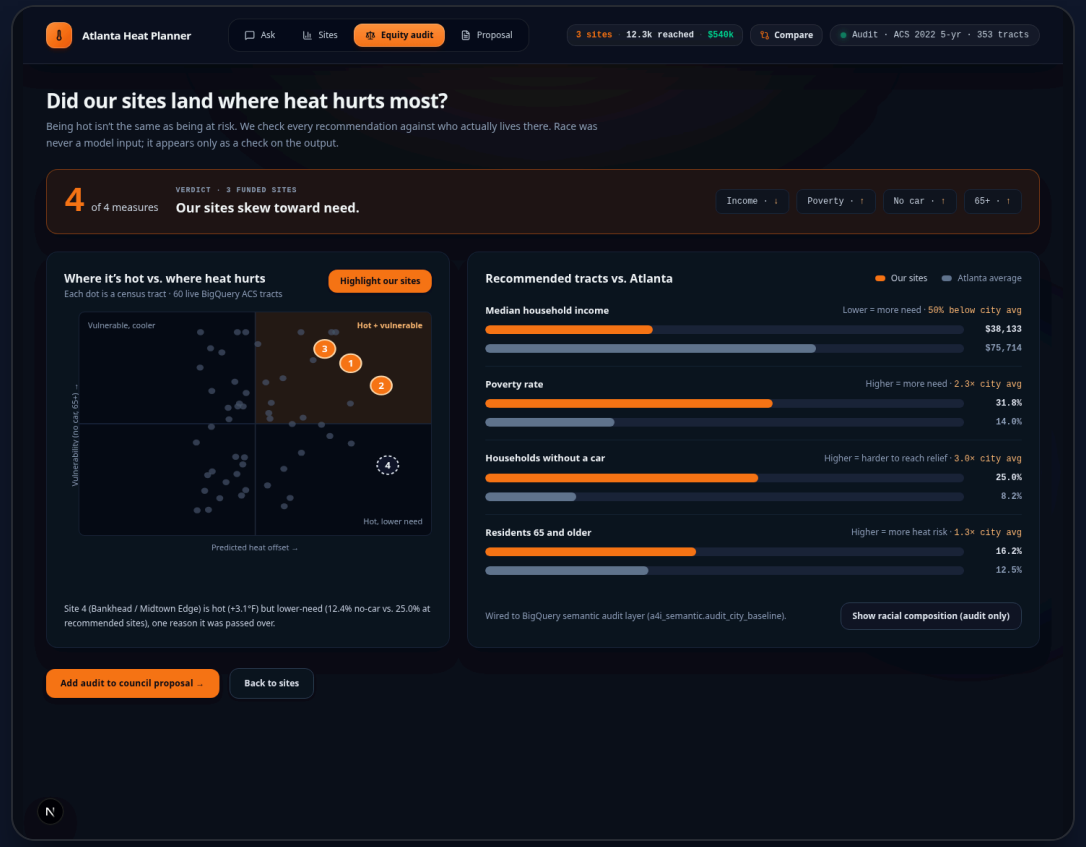


> ***Speaker Note:*** *"A random split on 1 Hz road readings leaks adjacent points and yields a fake 0.98 R^2. We aggregated to 150m geohash-7 cells, held out entire 5km geohash-5*

Equity Audit: 4 of 4 Measures Confirm Need-Based Prioritization

Post-Hoc Equity Audit vs. Atlanta Baseline (353 Tracts)

- **Strict Ethical Guardrail (DEC-005):** Race and ethnicity are never used in `a4i_core`, BQML training features, or site ranking. We audit recommended outputs post-hoc against `a4i_semantic.audit_city_baseline`.
- **Households Without a Vehicle: 25.0%** in 0.5-mi catchment (**50.41%** in core tracts) vs. **8.23%** Atlanta average → **3.0x to 6.12x higher transit dependency**.
- **Poverty Rate: 31.8%** in catchment (**44.05%** in core tracts) vs. **13.95%** Atlanta average → **2.3x to 3.16x higher poverty concentration**.
- **Seniors Aged 65+: 16.2%** in catchment (**19.44%** in core tracts) vs. **12.47%** Atlanta average → **1.3x to 1.56x higher senior vulnerability**.
- **Median Household Income: $38,133** in catchment (**$17,853** in core tracts) vs. **$75,714** Atlanta average (**50% to 76% below citywide median**).
- **Post-Hoc ACS Racial Check (Output Only):** Selected tracts are **90.81% Black residents** vs. 41.66% citywide (2.18x), proving physical heat + transit siting reaches historically redlined neighborhoods without demographic bias in the model.

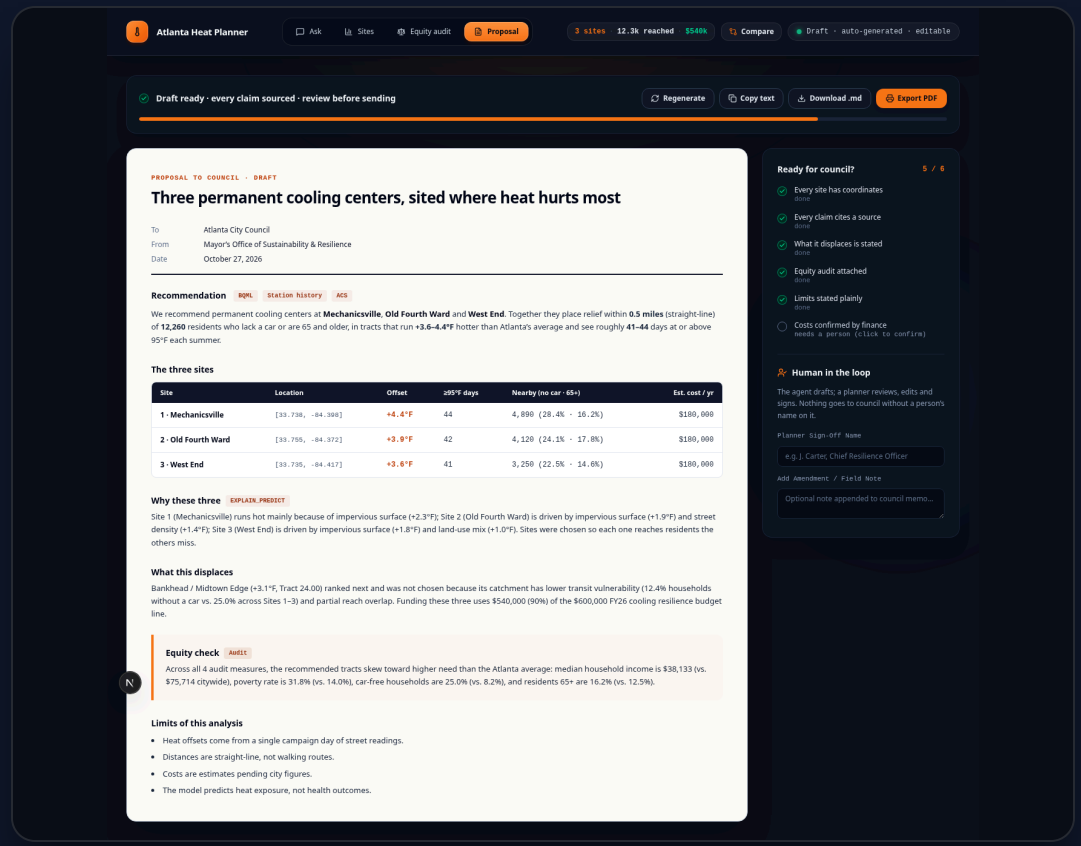


> ***Speaker Note:*** *"Race was never a model input. Yet when we run our post-hoc equity audit across all 353 Atlanta census tracts, our 3 recommended sites reach communities with 3.0x to 6.1x higher car-free households, 2.3x to 3.2x higher poverty, and half the citywide median income."*

Human-in-the-Loop Council Memo, Honest Limits & Live Links

Why Custom UI, Honest Boundaries & Submission Links

- **Why Custom Next.js UI on Cloud Run:** Planners defending a $600k vote need interactive Voronoi tract maps, Site 1 vs. Site 4 trade-off cards, and a **Human-in-the-Loop sign-off memo** (not a raw developer chat). Pinned `min-instances=1`.
- **Honest Refusal Gate (DEC-007):** `a4i_ml.point_features` enforces `in_footprint` (≤500 m from Heat Watch route). Queries in Marietta (12.3 km out) or Chicago are refused honestly.
- **Known Model Limitations:** (1) Single-day Heat Watch snapshot; (2) Predicts current exposure offset, cannot simulate cooling ΔT from planting new trees; (3) Straight-line 0.5-mi / 1.5-km buffers, not turn-by-turn walk isochrones; (4) 100% US Gov/CC0 data (zero OSM/ODbL).
- **Explore Qwiklabs Project:** qwiklabs-gcp-04-ec3596addc43
- **GitHub Repository:** https://github.com/chintamanenirc/A4I2026-challenge-1-walkability
- **Live Demo & Agent URLs:** urban-heat-ui · urban-heat-agent · heat-planner-api



> ***Speaker Note:*** *"Finally, the agent drafts a cited City Council memo, but nothing leaves without human planner sign-off. It states its limits plainly—single campaign day, straight-line buffers, and honest refusal outside the 500m footprint. Data: NOAA & CAPA Strategies Heat Watch (CC0), NOAA GHCN-Daily, US Census ACS. Thanks to the volunteers who drove the routes."*